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30 March 1971
SUPERSEDING
PA-PD-MI-2554, Rev. 0
10 July 1962

MILITARY SPECIFICATION
DELAY ELEMENT, CARTRIDGE: M45, ASSEMBLY

1. SCOPE

1.1 This specification covers Delay Element, Cartridge: M45, Assembly, for use in cartridges for propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-13739	- Composition, Delay
MIL-C-60539	- Coating, Anodic, Conventional for Aluminum and Aluminum Alloys

Standards

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions

Drawings

Munitions Command

D10522658	- Delay Element, Cartridge, M45, Assembly
C8826729	- Box, Packing, Ammunition for Delay Element, Cartridge, M45, Assembly
C8864162	- Ammunition Container, Metal, M21, for Delay Element, Cartridge, M45
IEL10522658	- Inspection Equipment List for Delay Element, Cartridge, M45, Assembly

FSC 1377

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(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y. 10016).

3. REQUIREMENTS

3.1 General - Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications.

3.2 Protective finish - Protective finish shall be in accordance with MIL-C-60539.

3.3 Vibration - The delay element assembly shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the primer or delay charge or derangement of components.

3.4 Primer sensitivity - The primer shall fire when drop tested at an energy level of 26.0 inch ounces and shall not fire when drop tested at an energy level of 4.0 inch ounces, in a delay element assembly void of pyrotechnical charge. Primer and firing plug area must be free of sealant.

3.5 Functional

3.5.1 The delay element shall not fail to fire.

3.5.2 Ballistics - (Delay Charge) - The delay charge shall comprise of increments of type II and type III formulations shown in MIL-C-13739 in the proportion and quantity fixed by the contractor to produce the delay time required in table I (See 6.2).

Table I

Temperature	Delay Time (Secs) (See 6.3)
-65° to -70° F	5.9 max.
65° to 75° F	5.2 max. 4.8 min.
160° to 165° F	3.9 min.

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3.6 Workmanship - Workmanship shall be in accordance with the requirements of the applicable drawings, referenced specifications and the following:

3.6.1 Metal parts - The metal parts shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.6.2 Burr - The metal part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.6.3 Foreign matter - The metal part or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance terms and definitions - Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection - Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample

4.2.1 Initial production sample - At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of: 23 anodized delay bodies, 23 anodized primer retainers for protective finish tests, 25 delay elements for all-fire primer sensitivity, 25 delay elements for no-fire primer sensitivity, 12 delay elements for vibration test (to be selected from ballistic sample) and 30 delay elements for ballistic tests, to be tested in accordance with table II and 10 sets of unassembled components (including explosive components) for dimensional and visual examination.

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Table II

Paragraphs Requirement	Test	Test	Sample Size	Type of Test Destr	Type of Test Non-Destr
3.2	4.4.1	Protective Finish	23 delay bodies 23 primer retainers	x x	- -
3.3	4.4.2	Vibration	12 delay elements (selected fr ballistic sample)	-	x
3.4	4.4.3	All-Fire Sensitivity	25 All-Fire Sensitivity	x	-
3.4	4.4.3	No-Fire Sensitivity	25 No-Fire Sensitivity	x	-
3.5.2	4.4.4	Ballistics	30 Delay Elements	x	-

The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.3.1.1 Examination and tests - After examination and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a government laboratory or such other facility as specified in the contract.

4.2.1.2 Ballistics - The sample for ballistic tests shall be temperature conditioned to determine compliance with table I. The sample for firing shall consist of the following:

Table III

Temperature	Not Vibrated	Vibrated
- 65° to - 70° F	6	4
65° to 75° F	6	4
160° to 165° F	6	4

4.2.1.3 Initial production sample failure - Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

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4.3 Inspection provisions

4.3.1 Lot

4.3.1.1 Submission of product - The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot - A lot shall consist of an identifiable group of metal parts assemblies produced by a homogeneous process and loaded with acceptable delay charges and primers from not more than one lot.

4.3.2 Examination - One hundred percent examination shall be performed for all critical defects. Examination for major or minor defects shall be performed on a defect basis in accordance with the classification of defects, using the sampling plan and associated criteria in tables IV and V. For an individual defect (single characteristic) the criteria in table IV shall be used. For total defects within a major or minor class, the criteria in table V shall be used. All non-conforming material shall be rejected. Four power (4X) magnification may be used to perform visual examination for such defects as burr and foreign matter. The roughness comparison specimens prescribed in ASA B46.1 shall be used as a basis of comparison for surface roughness determination.

Table IV

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% examination	-	-	-	-
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

Table V

Sample Size for Major Defects Number of Defects in Class	Total Allowable Defects Per Sample Size			
	110	150	225	300
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14
Sample Size for Minor Defects				
Number of Defects in Class	110	150	225	300
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

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4.3.2.1 Body, Delay, Drawing C8594081, covering a detail of Drawing D10522658

Defect	Method of Inspection
CRITICAL	
1. Diameter of flash hole	Gage
2. Metal defective	Visual
MAJOR	
101. Pitch diameter of thread, min.	Gage
102. Major diameter of thread, min.	Gage
103. Concentricity of body with pitch diameter of thread	Gage
104. Diameter of thread relief	Gage
105. Diameter of step at thread relief	Gage
106. Diameter of cavity, max.	Gage
107. Diameter of body, min.	Gage
108. Depth of cavity, max.	Gage
109. Length from cavity face to body diameter	Gage
110. Thread damaged or not full	Visual
111. Protective finish inadequate or defective	Visual
112. Burr	Visual
113. Foreign matter	Visual
MINOR	
201. Pitch diameter of thread, max.	Gage
202. Diameter of cavity, min.	Gage
203. Diameter of body, max.	Gage
204. Diameter of wrench holes	Gage
205. Diameter of thread pilot	Gage
206. Depth of cavity, min.	Gage
207. Depth of wrench holes	Gage
208. Total length	Gage
209. Length of thread pilot	Gage
210. Length of thread relief	Gage
211. Length of step at thread relief	Gage
212. Location of wrench holes	Gage
213. Radius or chamfer missing or incorrect	Visual
214. Surface finish improper	Visual

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4.3.2.2 Gasket, Aluminum, Drawing B8594082, covering a detail of Drawing D10522658

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Diameter, outside, min.	Gage
102. Diameter, inside, max.	Gage
103. Metal defective	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
201. Diameter, outside, max.	Gage
202. Diameter, inside, min.	Gage
203. Thickness	Gage
204. Surface finish improper	Visual

4.3.2.3 Plug, Firing, Drawing B8597234, covering a detail of Drawing D10522658

CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Concentricity of flange with body	Gage
102. Diameter of flange, min.	Gage
103. Diameter of body, min.	Gage
104. Radius of striker	Gage
105. Length of striker	Gage
106. Surface finish on body improper	Visual
107. Burr or sharp corner	Visual
108. Foreign matter	Visual
MINOR	
201. Diameter of flange, max.	Gage
202. Diameter of body, max.	Gage
203. Total length	Gage
204. Length of body	Gage
205. Radius missing or incorrect, except as otherwise classified	Visual
206. Surface finish improper, except as otherwise classified	Visual

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4.3.2.4 Primer, Percussion, M42, Drawing 8799925, covering a detail of Drawing D10522658

Defect	Method of Inspection
CRITICAL	
1. Improper assembly	Visual
2. Any part missing	Visual
3. Primer pellet broken, cracked or dislodged	Visual
4. Metal defective	Visual
5. Primer foil discolored	Visual
6. Mixed primers	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

4.3.2.5 Gasket, Drawing B10522395, covering a detail of Drawing D10522658

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Diameter, outside, min.	Gage
102. Diameter inside, max.	Gage
103. Durometer hardness	Gage
104. Material defective	Visual
105. Flash	Visual
106. Foreign matter	Visual
MINOR	
201. Diameter, outside, max.	Gage
202. Diameter, inside, min.	Gage
203. Thickness	Gage

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4.3.2.6 Cup, Retaining, Drawing D10522653, covering a detail of Drawing D10522658

	Defect	Method of Inspection
CRITICAL		
1.	Metal defective	Visual
MAJOR		
101.	Concentricity of flange with small hole	Gage
102.	Concentricity of body with small hole	Gage
103.	Concentricity of counterbore with small hole	Gage
104.	Diameter of flange, min.	Gage
105.	Diameter of body, min.	Gage
106.	Diameter of counterbore	Gage
107.	Diameter of small hole	Gage
108.	Depth of counterbore	Gage
109.	Surface finish of small hole, flange body and counterbore	Visual
110.	Burr or sharp edge	Visual
111.	Foreign matter	Visual
MINOR		
201.	Diameter of flange, max.	Gage
202.	Diameter of body, max.	Gage
203.	Total length	Gage
204.	Length of body	Gage
205.	Surface finish improper except as otherwise classified	Visual

4.3.2.7 Primer, Retainer, Drawing C10522657, covering a detail of Drawing D10522658

CRITICAL		
1.	Diameter of primer pocket	Gage
2.	Diameter of flash hole	Gage
3.	Depth of primer pocket	Gage
4.	Metal defective	Visual
MAJOR		
101.	Pitch diameter thread, max.	Gage
102.	Minor diameter thread, max.	Gage
103.	Concentricity flash hole with primer pocket	Gage
104.	Concentricity of flange with primer pocket	Gage
105.	Concentricity of large diameter with primer pocket	Gage
106.	Concentricity of large counterbore with primer pocket	Gage

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4.3.2.7 - Cont'd

MAJOR - Cont'd

	Defect	Method of Inspection
107.	Concentricity small counterbore with primer pocket	Gage
108.	Concentricity of body with primer pocket	Gage
109.	Concentricity of pitch diameter of thread with body	Gage
110.	Symmetry of flat with pitch diameter of thread	Gage
111.	Diameter of body min.	Gage
112.	Diameter small counterbore, max.	Gage
113.	Diameter large counterbore, max.	Gage
114.	Depth of small counterbore	Gage
115.	Depth to base of small counterbore	Gage
116.	Length of effective thread, min.	Gage
117.	Location of datum diameter	Gage
118.	Thread damaged or not full	Visual
119.	Protective finish inadequate or defective	Visual
120.	Surface finish of primer pocket and small counterbore, improper	Visual
121.	Burr or sharp edge	Visual
122.	Foreign matter	Visual

MINOR

201.	Pitch diameter of thread, min.	Gage
202.	Diameter of small counterbore, min.	Gage
203.	Diameter of large counterbore, min.	Gage
204.	Diameter of body, max.	Gage
205.	Diameter of flange	Gage
206.	Depth to bottom of threaded cavity	Gage
207.	Total length	Gage
208.	Length across wrench flat	Gage
209.	Length of wrench flat	Gage
210.	Location of flat	Gage
211.	Location of flange	Gage
212.	Radius or chamfer missing or incorrect	Visual
213.	Surface finish improper, except as otherwise classified	Visual

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4.3.2.8 Assembly, Drawing D10522658

Defect	Method of Inspection
CRITICAL	
1. Primer not seated to full depth of pocket	X-ray 1/ and Visual
2. Primer and firing plug contaminated with sealant	Visual
3. Improper assembly	X-ray 1/ and Visual
4. Any part missing	X-ray 1/ and Visual
5. Marking missing, incorrect or illegible	Visual
6. Gasket missing from firing plug	X-ray 1/ and Visual
MAJOR	
101. Concentricity of flange with body	Gage
102. Firing plug not flush or below retaining cup	Gage
103. Sealant between cup retaining and primer retainer, missing	Visual
MINOR	
201. Diameter of body	Gage
202. Total length, max.	Gage
203. Cement between gasket and primer retainer missing	Visual
204. Junction of primer and pocket not sealed and crimped	Visual

1/ The delay element shall be x-rayed with the element in a position horizontal to the rays. The x-ray properly identified shall be forwarded to the procuring activity.

4.3.2.9 Ammunition Container, Metal, M21 for Delay Element, Cartridge, M45, Assembly, Drawing C8864162

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual and Visual
202. Quantity of contents incorrect	Visual
203. Separator or filler missing	Visual
204. Heat seal incomplete, loose or distorted	Visual
205. Cut or damaged through impregnated box	Visual
206. Marking missing, incorrect or illegible	Visual

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4.3.2.10 Box, Packing, Ammunition for Delay Element, M45, Drawing C8826729

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual and Visual
202. Hardware missing, loose, broken or improperly engaged	Visual
203. Strapping missing, loose, broken or improperly assembled	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of cartons incorrect	Visual
208. Carseal missing, unsealed or improperly positioned	Visual

4.3.3 Inspection testing - The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Protective finish - Twenty-three units of each anodized component shall be selected for protective finish tests. Failure of any component to comply with the requirements of 3.2 shall be cause for rejection of the lot.

4.3.3.2 Vibration - Twelve of the 30 assembled x-rayed delay elements of the initial production sample shall be subjected to vibration. After vibration the 12 vibrated delay elements shall be x-rayed as directed in 1/ of 4.3.2.8. Failure of any unit to comply with the requirements of 3.3 shall be cause for rejection of the lot. When there is a change in the method of charging the delay element or a change in the lot of primers, 12 additional delay elements shall be selected and tested as indicated herein.

4.3.3.3 Primer sensitivity - Fifty primed delay elements from each lot shall be tested, using 25 cartridges for the all-fire test and 25 cartridges for the no-fire test. Failure of any primer to comply with the requirements of 3.4 shall be cause for rejection of the lot.

4.3.3.4 Functional - The following test shall be performed by the government:

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4.3.3.4.1 Ballistics - The test delay elements shall be temperature conditioned as indicated in table VI and functionally tested. Failure of any delay element to comply with the requirements of 3.5 shall be cause for rejection of the lot.

Table VI

Temperature	Sample
-65 to -70 F	10
65 to 75 F	10
160 to 165 F	10

4.3.3.5 Sealant application - The sample of delay elements used in the primer sensitivity test shall be disassembled and examined for proper sealant application. Failure of any delay element to comply with drawing requirements shall be cause for rejection of the lot.

4.3.4 Inspection equipment - The examination and tests shall be made using equipment listed on IEL 10522658.

4.4 Test methods and procedures

4.4.1 Protective finish - Test procedures for protective finish shall be in accordance with MIL-C-60539. All anodized components shall be sectioned longitudinally to better expose the critical surfaces to the salt spray.

4.4.2 Vibration - The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

4.4.3 Primer sensitivity - Remove the charged pyrotechnique body from the delay element assembly before conducting primer sensitivity test. The primer sensitivity test shall be made using an all-fire and a no-fire test. The all-fire drop test height shall be 11-½ inches and the no-fire drop height shall be .99 inches. One half of the prescribed sample of 50 cartridges shall be tested using a 1-3/16 inch diameter ball weighing 3.94 ± .02 ounces at each of the two heights, using a primer sensitivity machine equipped with a cartridge holder approved by the procuring activity. The head space shall be zero. All height measurements shall be made between the head of the firing pin and the bottom of the suspended ball. The primer sensitivity machine shall be checked to be sure the drop ball is impacting the center of the firing pin. All firing pin indent centers shall be within .020 of the center of the primer.

4.4.4 Functional

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4.4.4.1 Ballistics - The sample shall be grouped and temperature conditioned in accordance with 4.4.4.1.1 and rigidly supported in the test fixutre. The sample shall be tested with 5 minutes after removal from the conditioning chamber. When this time limit is exceeded the sample shall be reconditioned in accordance with 4.4.4.1.1.

4.4.4.1.1 Temperature conditioning - A standard inert M45 delay element containing a temperature sensitive element shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of 1 hour longer than the time necessary to bring the element contained in the dummy delay element to the required temperature. The vent holes of the delay element shall be sealed against the environmental atmosphere of the conditioning chamber during the conditioning period. Upon completion of temperature conditioning, the vent hole seal shall be removed.

4.4.5 Sealant application - Upon completion of the primer sensitivity test, each delay element will be disassembled, exposing the firing plug and primer areas for examination.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking - Packing and marking shall be in accordance with Drawings C8826729 and C8864162, except that the marking shall refer to Delay Element, Cartridge, M45 Assembly.

6. NOTES

6.1 Ordering data - Procurement document should specify the title, number and date of this specification.

6.2 Advisory delay charge composition (formulation) - Consists of 3 increments of charge as follows: Approximately 5 grains Type II, 13½ grains Type III and 5 grains, Type II, charged in that

6.3 Delay time is defined as the time lapse between the impact of the firing pin on the primer and evidence of gas flow from the vent hole of the delay element.

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6.4 Supersession data - This specification includes the requirements of Purchase Description FA-PD-MI-2554 dated 10 July 1962.

Custodian:

Army - MU

Preparing activity:

Army -MU

Project No. 1377-A224

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

5. SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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